

# One-minute Concept Animation

Storyboard and supporting documentation for the 60-second film.

# 12

OF 12 UPLOAD SLOTS

**15,000 m<sup>2</sup>**

SITE AREA

**44.5% → 64.6%**

COMFORTABLE DAYLIGHT HOURS

**−7.13 °C**

HEAT INDEX UNDER CANOPY

**131**

TREES PLANTED

▶ VISIT THE LIVE PROJECT PORTAL

[wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

01

## WHAT IS IN THIS FILE

Concept Animation Storyboard

DRAWING

Fig10 masterplan

PLAN

02

## THE SCALE OF THE ANALYSIS BEHIND IT

**39 years**

of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

**8,760 hours**

of sun position ray-traced against 131 trees for every hour of the year

**15,000**

one-metre ground cells modelled for shade and comfort, site-wide

**4 models**

trained and tested for leakage — two of them decide what this document reports

**41 checks**

run on every rebuild, so a wrong number fails loudly instead of shipping quietly

## VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository [github.com/wasim437/AI\\_SAFA](https://github.com/wasim437/AI_SAFA)

Live portal [wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

Produced by **BUILD** [tools/sync\\_film.py](#) **TESTS** [tests/test\\_film.js](#)

Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

# A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

12

OF 12

## 01 THE TEN PHASES

### P1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

### P3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

### P5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m<sup>2</sup>

### P7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

### P9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

THIS DOCUMENT

### P2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

### P4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

### P6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

### P8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

### P10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

## 02 THE CODE AND DATA BEHIND THIS DOCUMENT

### Sync film

tools

### Test film

tests

## 03 REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

UPLOAD SLOT 12 OF 12 · ONE-MINUTE CONCEPT ANIMATION — STORYBOARD

# Concept Animation — Storyboard

Sixty seconds, drawn from the project's own analysis

The film is not an illustration of the design; it is rendered from the design. Its geometry comes from the same arc definition as the masterplan and its sun from the same 8,760-hour solar model, so it cannot drift away from the drawings the way an animation produced separately would.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m<sup>2</sup> · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 03 August 2026 · every quantity regenerated by `python run_analysis.py`

## 1. Structure — five scenes, sixty seconds

| Time    | Scene               | What it shows                                                                                                                                                         |
|---------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–7 s   | <b>The site</b>     | Al Safa 2 as it is today: flat, exposed, and empty at midday                                                                                                          |
| 7–18 s  | <b>The heat</b>     | The reason it is empty — the modelled heat index across a summer day, and the hours that are lost to it                                                               |
| 18–38 s | <b>The crescent</b> | The arc builds and plants itself while a full computed day passes over it. 131 trees, the falaj, the radial alleys and the rooms appear in the order they are reached |
| 38–48 s | <b>Beneath it</b>   | Eye level on the walk: the dappled soffit, the louvre cutting the low sun, the channel at the edge                                                                    |
| 48–60 s | <b>The proof</b>    | The measured result — comfortable hours rising from 44.5% to 64.6% — as the park fills at dusk                                                                        |

Scene boundaries are asserted by `tests/test_film.js`, which renders every frame and fails on any gap, overlap or invalid value.

## 2. Why it can be trusted against the drawings

- The arc is the same 141 m radius as the masterplan, regenerated by `tools/sync_film.py` from `src/plan.py`.
- The sun positions are the same NREL-computed values used by the shade model — the shadows move correctly for the site's latitude and the time of year being shown.
- The tree count, the 0.9 m channel and the room layout are read from the same geometry as every drawing in this submission.
- Every frame is rendered by an automated test that fails on any NaN or undefined value, so a geometry change cannot break the film quietly.

## 3. How to produce the video file

- Open `concept_film.html` full-screen in a browser.
- Start a screen recording (Win + Alt + R on Windows, Shift + Cmd + 5 on macOS).
- Press Restart and let all sixty seconds run.
- Stop, trim the ends, and add a voiceover if wanted.

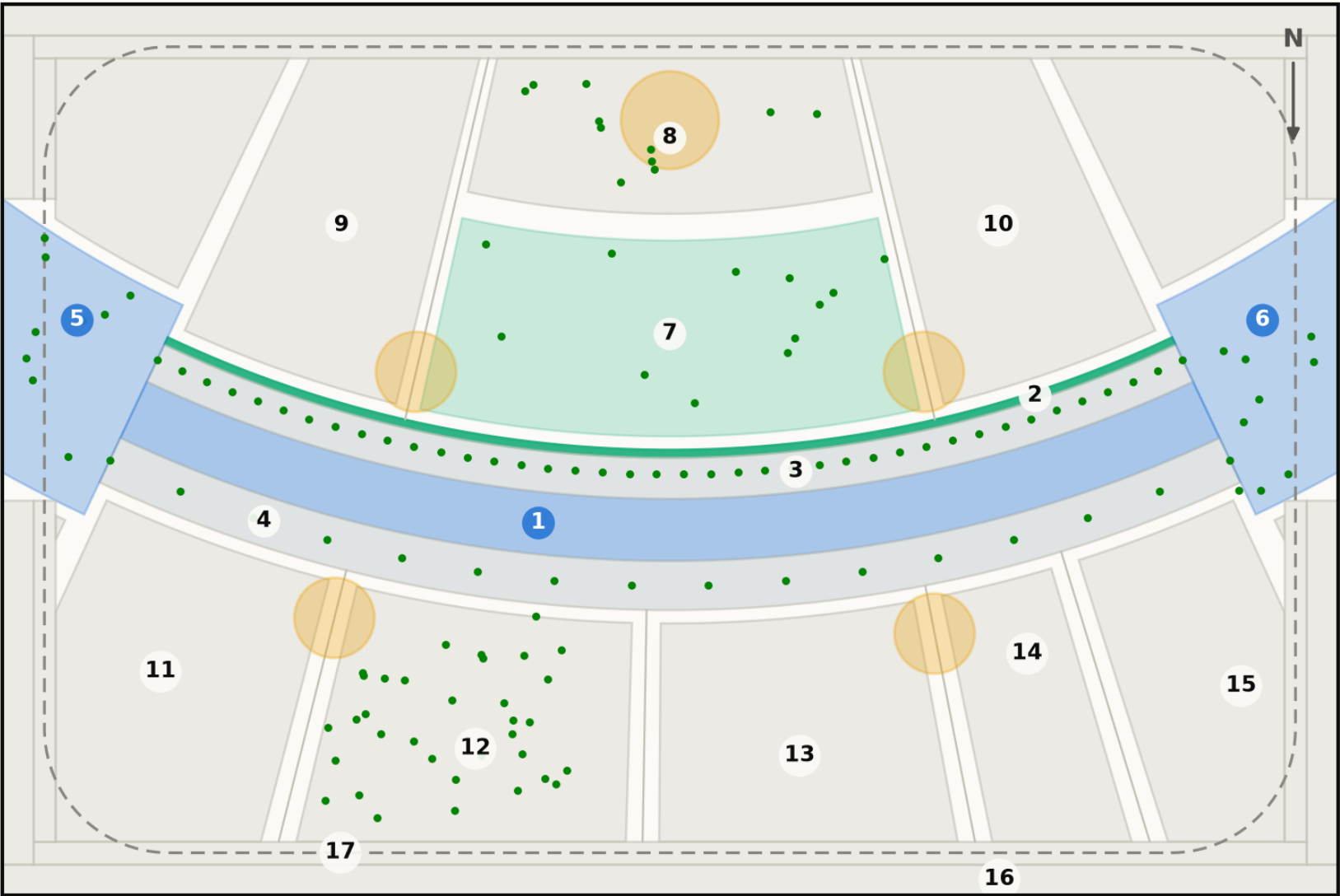
The film shows the **analysed** scheme exactly: the same arc, the same trees, the same solar model. It is therefore consistent with every number quoted elsewhere in this submission.

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



20 m

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.  
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

ROOM SCHEDULE

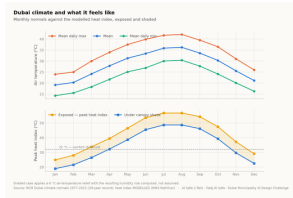
|                                      |                                     |           |
|--------------------------------------|-------------------------------------|-----------|
| 1                                    | Al Mamsha — the Crescent Walk       | 871 m²    |
| 2                                    | Al Falaj — the water channel        | 105 m²    |
| 3                                    | Crescent Shade Margin (N)           | 549 m²    |
| 4                                    | Crescent Shade Margin (S)           | 715 m²    |
| 5                                    | West Gate Majlis                    | 439 m²    |
| 6                                    | East Gate Majlis                    | 439 m²    |
| 7                                    | Al Nakhil — the Oasis Basin         | 1,140 m²  |
| 8                                    | Quiet Contemplation Garden          | 707 m²    |
| 9                                    | Children's Dune Play                | 1,267 m²  |
| 10                                   | Family Picnic Grove                 | 1,267 m²  |
| 11                                   | Outdoor Fitness Terrace             | 908 m²    |
| 12                                   | Native Planting / Biodiversity Wadi | 894 m²    |
| 13                                   | Community Plaza & Event Lawn        | 787 m²    |
| 14                                   | Souk Kiosks & Services              | 417 m²    |
| 15                                   | Multipurpose Sports Lawn            | 630 m²    |
| 16                                   | Al Kathib — the dune berm           | 1,464 m²  |
| 17                                   | Al Madar — the perimeter loop       | 986 m²    |
| Al Sikkak — shaded alleys & setbacks |                                     | 1,414 m²  |
| Total                                |                                     | 15,000 m² |

# Every drawing and every chart in this project

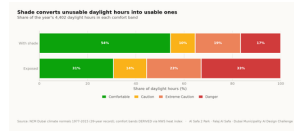
12

OF 12

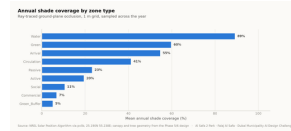
All 18 images the pipeline produces, whichever slot they belong to — each labelled with its class and linked to the full-resolution original. Nothing here is hand-drawn.



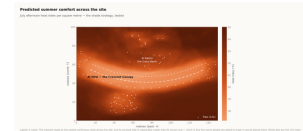
**Fig01 climate and comfort**  
analysis output



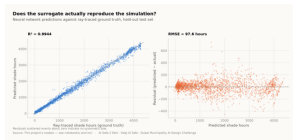
**Fig02 comfort bands**  
analysis output



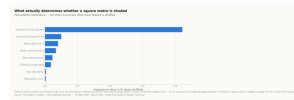
**Fig03 shade by zone**  
analysis output



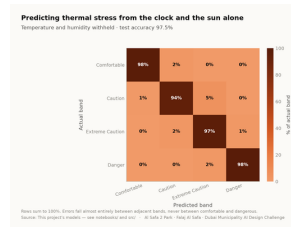
**Fig04 site comfort map**  
analysis output



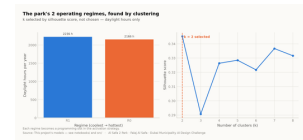
**Fig05 surrogate performanc**  
analysis output



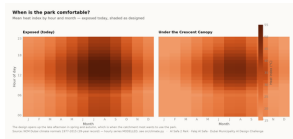
**Fig06 feature importance**  
analysis output



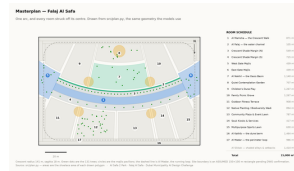
**Fig07 confusion matrix**  
analysis output



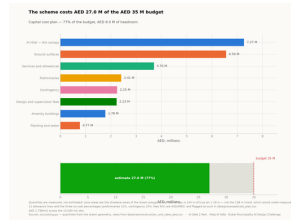
**Fig08 microclimate regimes**  
analysis output



**Fig09 diurnal comfort**  
analysis output



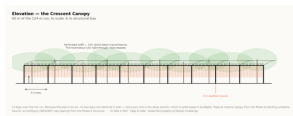
**Fig10 masterplan**  
analysis output



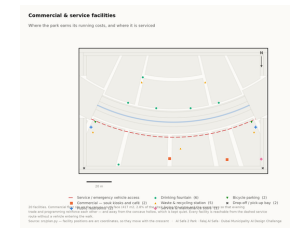
**Fig11 cost plan**  
analysis output



**Circulation**  
technical drawing



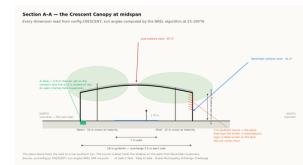
**Elevation**  
technical drawing



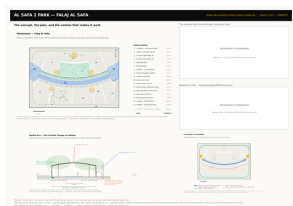
**Facilities**  
technical drawing



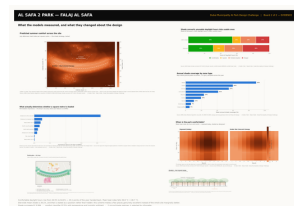
**Planting**  
technical drawing



**Section**  
technical drawing



**Board 1 concept**  
presentation board



**Board 2 evidence**  
presentation board

# Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

## MEASURED RESULT

models/headline\_metrics.json

|                                          |         |
|------------------------------------------|---------|
| Annual daylight hours modelled           | 4,402   |
| Comfortable daylight hours — today       | 44.5%   |
| Comfortable daylight hours — as designed | 64.6%   |
| Mean heat-index reduction under canopy   | 7.13 °C |
| Peak heat index, exposed                 | 56.8 °C |
| Peak heat index, shaded                  | 48.7 °C |
| Crescent Walk shaded, canopy and louvre  | 87.3%   |
| Site-wide mean shade                     | 34.1%   |
| Trees planted                            | 131     |

## TRAINED MODELS, ON HELD-OUT TEST SETS

models/model\_metrics.json

|                                                              |                        |
|--------------------------------------------------------------|------------------------|
| M1a Random Forest — shade surrogate, test R <sup>2</sup>     | 0.9982                 |
| M1b Neural network — deployed surrogate, test R <sup>2</sup> | 0.9944                 |
| M2 Gradient Boosting — comfort band, test accuracy           | 97.5%                  |
| M3 K-Means — regimes selected by silhouette                  | k = 2                  |
| Training / validation / test split                           | 10,500 / 2,250 / 2,250 |
| Random seed, fixed for reproducibility                       | 42                     |

## WHAT THE CHECKS PROTECT

each one asserts what would otherwise drift silently

### Climate fidelity

the modelled year reproduces the published normals it was built from

### No overlaps

no room in the schedule overlaps another

### Areas close

every area sums back to 15,000 m<sup>2</sup>

### Budget held

the cost plan stays inside AED 35 M

### No leakage

no model can see its own answer in its inputs

## RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js    # 64 portal checks
node tests/test_film.js          # every frame of the film
```

# The complete project, and where to check it

Every one of the 120 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository [github.com/wasim437/AI\\_SAFA](https://github.com/wasim437/AI_SAFA)

Live portal [wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

## THE TWELVE UPLOAD FILES (12)

|                                                                                              |                                              |
|----------------------------------------------------------------------------------------------|----------------------------------------------|
| <a href="#">UPLOAD THESE 12 FILES/01_Design_Narrative_and_Concept.pdf</a>                    | Design Narrative & Concept                   |
| <a href="#">UPLOAD THESE 12 FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf</a> | Neighborhood Park Preliminary Design Ma...   |
| <a href="#">UPLOAD THESE 12 FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf</a> | Concept Plans and Spatial Organization Di... |
| <a href="#">UPLOAD THESE 12 FILES/04_Key_Sections_and_Elevations.pdf</a>                     | Key Sections & Elevations                    |
| <a href="#">UPLOAD THESE 12 FILES/05_3D_and_Spatial_Visualizations.pdf</a>                   | 3D & Spatial Visualizations                  |
| <a href="#">UPLOAD THESE 12 FILES/06_AI_Methodology_Report.pdf</a>                           | AI Methodology Report                        |
| <a href="#">UPLOAD THESE 12 FILES/07_User_Experience_and_Activation_Strategy.pdf</a>         | User Experience & Activation Strategy        |
| <a href="#">UPLOAD THESE 12 FILES/08_Sustainability_Concept_and_Strategy.pdf</a>             | Sustainability Concept & Strategy            |
| <a href="#">UPLOAD THESE 12 FILES/09_Material_and_Landscape_Palette.pdf</a>                  | Material & Landscape Palette                 |
| <a href="#">UPLOAD THESE 12 FILES/10_Complete_Design_Report.pdf</a>                          | Complete Design Report                       |
| <a href="#">UPLOAD THESE 12 FILES/11_Site_Analysis_and_Human-Centric_Research.pdf</a>        | Site Analysis & Human-Centric Research       |
| <a href="#">UPLOAD THESE 12 FILES/12_One-minute_Concept_Animation.pdf</a>                    | One-minute Concept Animation                 |

## PROJECT DOCUMENTS (7)

|                                                   |                                                |
|---------------------------------------------------|------------------------------------------------|
| <a href="#">AL_SAFA_MASTER_PROMPT.md</a>          | The whole design, stated for visualisation     |
| <a href="#">DATA_SOURCES.md</a>                   | Every source, its period, and its limitations  |
| <a href="#">LINKS.md</a>                          | Every public URL this submission points at     |
| <a href="#">PROJECT_PLAN.md</a>                   | Requirements, phases, status, and what is left |
| <a href="#">PUSH_TO_GITHUB.md</a>                 | Project document                               |
| <a href="#">README.md</a>                         | The design argument, written for a juror       |
| <a href="#">EXPLAIN_THE_PROJECT/START_HERE.md</a> | The project in plain language                  |

## THE WRITTEN REPORTS (11)

|                                                                              |                                  |
|------------------------------------------------------------------------------|----------------------------------|
| <a href="#">reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf</a>        | Generated from the live analysis |
| <a href="#">reports/pdf/Concept_Animation_Storyboard.pdf</a>                 | Generated from the live analysis |
| <a href="#">reports/pdf/Phase2_Problem_Definition_Report.pdf</a>             | Generated from the live analysis |
| <a href="#">reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf</a>     | Generated from the live analysis |
| <a href="#">reports/pdf/Phase4_Concept_Development_Report.pdf</a>            | Generated from the live analysis |
| <a href="#">reports/pdf/Phase5_Masterplan_Development_Report.pdf</a>         | Generated from the live analysis |
| <a href="#">reports/pdf/Phase6_Detailed_Design_Report.pdf</a>                | Generated from the live analysis |
| <a href="#">reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf</a> | Generated from the live analysis |
| <a href="#">reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf</a> | Generated from the live analysis |
| <a href="#">reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf</a>  | Generated from the live analysis |
| <a href="#">reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf</a>  | Generated from the live analysis |

## THE ANALYSIS FIGURES (11)

|                                                         |                                              |
|---------------------------------------------------------|----------------------------------------------|
| <a href="#">figures/fig01_climate_and_comfort.png</a>   | Analysis output — computed from project data |
| <a href="#">figures/fig02_comfort_bands.png</a>         | Analysis output — computed from project data |
| <a href="#">figures/fig03_shade_by_zone.png</a>         | Analysis output — computed from project data |
| <a href="#">figures/fig04_site_comfort_map.png</a>      | Analysis output — computed from project data |
| <a href="#">figures/fig05_surrogate_performance.png</a> | Analysis output — computed from project data |
| <a href="#">figures/fig06_feature_importance.png</a>    | Analysis output — computed from project data |
| <a href="#">figures/fig07_confusion_matrix.png</a>      | Analysis output — computed from project data |
| <a href="#">figures/fig08_microclimate_regimes.png</a>  | Analysis output — computed from project data |
| <a href="#">figures/fig09_diurnal_comfort.png</a>       | Analysis output — computed from project data |
| <a href="#">figures/fig10_masterplan.png</a>            | Analysis output — computed from project data |

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.



figures/fig11\_cost\_plan.png

Analysis output — computed from project data

THE TECHNICAL DRAWINGS AND BOARDS (7)

|                                         |                                                |
|-----------------------------------------|------------------------------------------------|
| design/visuals/circulation_crescent.png | Technical drawing — generated from src/plan.py |
| design/visuals/elevation_crescent.png   | Technical drawing — generated from src/plan.py |
| design/visuals/facilities_crescent.png  | Technical drawing — generated from src/plan.py |
| design/visuals/planting_crescent.png    | Technical drawing — generated from src/plan.py |
| design/visuals/section_crescent.png     | Technical drawing — generated from src/plan.py |
| design/boards/board_1_concept.png       | Technical drawing — generated from src/plan.py |
| design/boards/board_2_evidence.png      | Technical drawing — generated from src/plan.py |

THE ANALYSIS CODE (13)

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| src/__init__.py | Analysis module                                       |
| src/boards.py   | The two presentation boards                           |
| src/climate.py  | The 8,760-hour year rebuilt from NCM normals          |
| src/config.py   | Every constant the project reads                      |
| src/costing.py  | The cost model against the AED 35 M ceiling           |
| src/dataset.py  | Assembles the ML training tables                      |
| src/drawings.py | Section, elevation, circulation, planting, facilities |
| src/figures.py  | The analysis charts                                   |
| src/models.py   | The four models                                       |
| src/plan.py     | SINGLE SOURCE of the crescent geometry                |
| src/solar.py    | Sun position and shadow ray-tracing                   |
| src/viz.py      | Analysis module                                       |
| run_analysis.py | Runs the whole pipeline end to end                    |

THE BUILD AND SYNC TOOLS (11)

tools/\_\_init\_\_.py  
tools/build\_docs.py  
tools/build\_links.py  
tools/build\_notebook.py  
tools/build\_reports.py  
tools/build\_submission\_pdfs.py  
tools/organize\_repo.py  
tools/report\_content.py  
tools/sync\_film.py  
tools/sync\_portal.py  
tools/sync\_submission.py

THE DATA, AS ISSUED (7)

|                                          |                                       |
|------------------------------------------|---------------------------------------|
| data/raw/construction_unit_rates_aed.csv | Source dataset                        |
| data/raw/dubai_climate_normals_ncm.csv   | Source dataset                        |
| data/raw/dubai_population_dsc_2023.csv   | Source dataset                        |
| data/raw/site_zoning_schedule.csv        | The measured room schedule            |
| data/raw/sources.json                    | Every source dataset, with its period |
| data/raw/species_water_carbon_rates.csv  | Source dataset                        |
| data/raw/walk_catchment_rings.csv        | Source dataset                        |

THE DATA, AS PROCESSED (6)

|                                                |                                           |
|------------------------------------------------|-------------------------------------------|
| data/processed/cost_plan.csv                   | The capital cost plan, line by line       |
| data/processed/hourly_climate_comfort_8760.csv | The 8,760-hour climate and comfort series |
| data/processed/masterplan_geometry.json        | The plan geometry, as exported            |
| data/processed/planting_layout.csv             | Every one of the 131 trees                |
| data/processed/schema.json                     | Generated by the pipeline                 |

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

# The complete project — continued

[data/processed/spatial\\_grid\\_comfort.csv](#)

The 15,000-cell ground grid

## THE TRAINED MODELS (3)

[models/cost\\_summary.json](#)

Model output

[models/headline\\_metrics.json](#)

The headline numbers

[models/model\\_metrics.json](#)

Trained-model metrics

## THE TESTS (2)

[tests/test\\_pipeline.py](#)

41 correctness checks

[tests/test\\_film.js](#)

Every frame of the concept film

## THE NOTEBOOK (1)

[notebooks/AL\\_SAF\\_A\\_2\\_PARK\\_COMPLETE\\_ANALYSIS.ipynb](#)

The complete analysis, outputs embedded

## THE WEBSITE AND ANALYTICS PORTAL (9)

[docs/index.html](#)

The project website

[docs/SUBMISSION\\_GUIDE.md](#)

Published site

[docs/\\_PORTAL/gallery\\_data.js](#)

Published site

[docs/\\_PORTAL/plan\\_geometry.js](#)

Published site

[docs/\\_PORTAL/portal.js](#)

Published site

[docs/\\_PORTAL/portal\\_data.js](#)

Published site

[docs/\\_PORTAL/selftest.js](#)

Published site

[docs/\\_PORTAL/DATA\\_AUDIT.md](#)

Published site

[docs/\\_PORTAL/README.md](#)

Published site

## THE COMPETITION BRIEF, AS ISSUED (7)

[00\\_BRIEF/Ai Park - Master Plan \(4\).jpg](#)

Dubai Municipality source document

[00\\_BRIEF/Ai Park Scope of Work & General Terms & Conditions \(5\).pdf](#)

Dubai Municipality source document

[00\\_BRIEF/Ai Safa Park 2 Plan \(5\).dwg](#)

Dubai Municipality source document

[00\\_BRIEF/MAIN WEBSITE DETAILS.txt](#)

Dubai Municipality source document

[00\\_BRIEF/Neighborhood Parks Manual \(4\).pdf](#)

Dubai Municipality source document

[00\\_BRIEF/NEIGHBORHOOD PARKS MANUAL\\_TEXT.txt](#)

Dubai Municipality source document

[00\\_BRIEF/SCOPE\\_OF\\_WORK\\_FULL\\_TEXT.txt](#)

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## THE TWELVE SUBMISSION FOLDERS (12)

[submission/01\\_Design\\_Narrative\\_Concept/MANIFEST.md](#)

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[submission/02\\_Preliminary\\_Design\\_Masterplan/MANIFEST.md](#)

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[submission/03\\_Concept\\_Plans\\_Spatial\\_Diagrams/MANIFEST.md](#)

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[submission/06\\_AI\\_Methodology\\_Report/MANIFEST.md](#)

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[submission/07\\_User\\_Experience\\_Activation\\_Strategy/MANIFEST.md](#)

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[submission/10\\_Complete\\_Design\\_Report/MANIFEST.md](#)

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[submission/12\\_Concept\\_Animation\\_Video/MANIFEST.md](#)

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## WORKING HISTORY (1)

[archive/withdrawn\\_visuals/README.md](#)

Images withdrawn on purpose, and why

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.